Vario USB: A Reproducible Bench Characterization of Barometric and Barometric-IMU Vertical-Velocity Estimation for a Low-Cost Variometer

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Repository: github.com/raghavmalik89/vario-usb-workbook · Commit series b3030bb–bc2f3f2

Abstract

We present a bench characterization of vertical-velocity (vario) estimation for a low-cost USB variometer built from two MS5611 barometric clones and an MPU6050 inertial measurement unit on an ESP32-S3. A causal, timestamp-driven replay engine re-runs 29 filter configurations over 22 committed captures (quiet and office statics, elevator rides, drops, tosses, controlled hand motion), with whole-session separation between tuning and validation, physically equivalent 45/90 Hz parameterizations, and all latency figures reported as derived proxies rather than ground truth. Measured findings: raw barometric differentiation is unusable (9.0–22.1 m/s RMS at rest), the pressure ADC word dominates sensor noise (2.56–2.89 Pa RMS, equivalent to 0.21–0.24 m of altitude), and a constant-velocity Kalman filter reaches a rest vertical-velocity floor of 0.10–0.11 m/s in a quiet house — seven times better than the same estimator in a drafty office (0.71–0.80 m/s), demonstrating that ambient pressure drafts, not the sensor, set the practical noise floor. Real-IMU fusion equals baro-only estimation at rest and on the elevator but degrades it 1.4–2.2x under tumble because a 6-DOF attitude estimator without magnetometer leaves gravity residuals of 6.5–10.3 m/s²; simulation shows that with known attitude the same fusion architecture reduces climb-tracking error 50x and coasts through a 1 s barometer blackout at 0.034 m/s RMS. Audio-policy evaluation exposes a missed-lift defect (18.9–21.6% of lifts) in the otherwise most accurate robust-adaptive filter, and a freeze defect in a dual-timescale filter is root-caused to a self-referencing replace step. No final estimator is selected; retained candidates, limitations, and reproducibility references are given.

1. Introduction

A variometer is the central flight instrument of soaring: it reports vertical velocity so the pilot can centre thermals and detect sink. The core engineering tension is latency versus noise. Barometric altitude is available at tens of hertz with sub-metre resolution, but differentiating altitude to velocity amplifies noise roughly with bandwidth; any low-pass or estimator that suppresses noise also delays the onset and release of a lift, and pilots report release detection (exit from lift) as being at least as important as onset. Consumer variometers additionally contain IMUs, and open-source designs (e.g. har-in-air's ESP32 vario) fuse barometer and accelerometer; how much of that benefit survives on real low-cost hardware with consumer-grade IMUs is rarely quantified.

This paper documents a bench programme that treats the variometer as an estimation problem to be measured, not assumed. All experiments run on one physical rig whose raw captures are committed to version control, so every figure in this paper is reproducible from the repository by command-line invocation. The contributions are:

1) A causal replay engine that re-runs filters against recorded raw ADC streams with physically derived sample intervals, preserves corrupt frames and missing-sample events in diagnostics, and keeps sensor streams independent unless an explicit fusion algorithm is being evaluated (Section 3).

2) A characterization of two GY-63 MS5611 clones and a genuine MPU6050 on one I2C rig, including clone scale bias and the dominance of the pressure ADC word in sensor noise (Section 4.1).

3) A 29-filter comparison across five families (window smoothers, frequency-domain low-pass, trackers, Kalman variants, robust/gated estimators) with whole-session tuning/validation separation, physically equivalent 45/90 Hz parameterizations, and CPU-cost measurements relevant to an embedded target (Sections 4.3–4.4).

4) A real-IMU fusion validation separating the Kalman core from the attitude estimator, showing that attitude quality — not the filter — is the fusion bottleneck on this hardware (Section 4.5), supported by a simulation ceiling with known attitude (Section 4.7).

5) An audio-behaviour evaluation covering lift onset, exit-from-lift, false beeping, and a missed-lift defect in the most accurate filter (Section 4.4), plus fault-injection and C99-portability results (Section 4.6).

Throughout, claims are tagged [M] measured, [S] simulated, [D] derived proxy, [H] hypothesis, or [U] unverified. No final estimator is selected in this paper; the programme deliberately ends at a shortlist with measured trade-offs.

2. Background and Related Work

2.1 Barometric altimetry

Barometric altitude follows from the pressure-altitude relation of the 1976 ICAO standard atmosphere as tabulated in the MS5611 datasheet [1]:

$$h(p) = 44330\left[1 - \left(\frac{p}{p_0}\right)^{0.1903}\right]$$

with p the measured pressure and p0 = 1013.25 hPa the sea-level reference. The MS5611-01BA03 is a 24-bit barometric module; at oversampling ratio (OSR) 4096 its datasheet RMS noise is 0.012 mbar typical [1]. Stochastic noise modelling for barometric altimeters is treated by Sabatini and Genovese [6]. Vertical velocity is the derivative of h; direct differencing is the trivial estimator and, as our measurements confirm (Section 4.3), the worst.

2.2 Recursive estimation

The classical framework is the Kalman filter [3]; Welch and Bishop give the standard formulation [4]. The workhorse model here is a two-state constant-velocity (CV) filter over altitude h and velocity v with process spectral density q and measurement variance r:

$$F = [[1,\ dt],\ [0,\ 1]],\quad Q = q\,[[dt^3/3,\ dt^2/2],\ [dt^2/2,\ dt]],\quad H = [[1,\ 0]]$$

In fusion mode each timestamped accelerometer sample propagates the state through the gravity-compensated Earth-vertical acceleration a_z - g:

$$x_{k+1} = F\,x_k + [dt^2/2,\ dt]^T (a_z - g),\quad a_z = (R(q_k)\,a_{meas})_z$$

2.3 Attitude estimation

Gravity compensation requires orientation. We use the gradient-descent IMU/MARG orientation filter of Madgwick, Harrison and Vaidyanathan [5], which fuses accelerometer and gyroscope at low computational cost suitable for microcontrollers; the quaternion update integrates gyro rates with a feedback term driving the measured direction of gravity toward the model direction:

$$q_{k+1} = q_k + \tfrac{\Delta t}{2}\, q_k \otimes \left[0,\ \omega_k - \hat{b}_k\right]$$

The rig has no magnetometer, so yaw is unobservable in principle and pitch/roll are only as good as the acceleration reference — a limitation that Section 4.5 quantifies under tumble.

2.4 Noise stability analysis

Allan deviation [7] separates white noise from drift by averaging successive phase averages; we report it on detrended altitude at tau = 1 s and 10 s as a stability fingerprint of each capture:

$$\sigma_y^2(\tau) = \tfrac{1}{2}\left\langle (\bar{y}_{k+1} - \bar{y}_k)^2 \right\rangle$$

2.5 Open-source practice

The closest open references are har-in-air's ESP32 IMU/BARO/GPS variometer [8], which couples a Kalman altitude estimator with an IMU and is the de facto design pattern for DIY variometers, and Gallostra's AltitudeEstimation library [9]. Audio conventions (rising tone with climb rate, onset/exit thresholds) follow XCSoar's established glide-computer practice [10]. This paper differs from both in that every design claim is tied to a committed raw capture and a replayable metric rather than to field impressions.

3. Methodology

3.1 Hardware rig

One ESP32-S3 development board hosts two independent I2C buses. Bus A (GPIO4/5) carries an MPU6050 IMU at 0x69 and an MS5611 clone at 0x76; bus B (GPIO6/7) carries a second MS5611 clone at 0x77. The GY-63 modules are low-cost MS5611 breakouts from the same supplier batch; clone identity is verified by PROM CRC and PROM[0] words (0x0878 / 0x0884) [M]. The IMU is a genuine MPU6050 (WHO_AM_I 0x68). All orientation-sensitive firmware assumes the X-up mounting of the rig [M]. Table 1 summarizes the as-built.

Table 1. *As-built rig. A third barometer on a bit-banged bus C was probed during bring-up but is not part of the dataset.*

Device	Bus	Address	Role	Verified by
MS5611 clone #1 (GY-63)	A (GPIO4/5)	0x76	Barometer sid 0	PROM[0]=0x0878, CRC ok
MPU6050 (genuine)	A (GPIO4/5)	0x69	IMU, 200 Hz accel+gyro	WHO_AM_I=0x68
MS5611 clone #2 (GY-63)	B (GPIO6/7)	0x77	Barometer sid 1	PROM[0]=0x0884, CRC ok

3.2 Capture protocol and dataset

Captures are raw telemetry frames (P/T/E lines: pressure ADC, temperature ADC, error codes) with hardware timestamps, recorded over USB at 921600 baud using a fixed command-line tool. Twenty-two captures are committed: bench statics at two firmware generations (t001–t003), elevator and step-drop (t004), office statics plus elevator, drop, toss and controlled-move sessions (t005–t006), the same set with the IMU streamed at 200 Hz (t007), and a quiet-house static (t008). Data-quality diagnostics (sequence drops, corrupt frames) accompany every capture; the t005–t008 motion captures used here were clean (0 sequence drops at ingest). Capture metadata (OSR, rate, label) is stored in JSON sidecars next to each CSV, and the benchmark regenerates its reports from these files only.

3.3 Causal replay engine

The benchmark never re-simulates noise: it replays recorded ADC streams through each filter with sampling intervals derived from physical timestamps (never assumed nominal), handling irregular gaps, spikes and dropouts exactly where they occurred. Corrupt frames and missing-sample events are preserved in per-run diagnostics rather than silently dropped. Individual sensor streams are kept independent — the two barometers are never averaged — except when an explicitly defined fusion algorithm is under test, so that cross-sensor noise structure (shared chamber pressure vs independent electronics) remains observable in the results. An event detector derives lift events from the reference stream of each session; onset/release latencies are then computed per filter against those derived references and are reported as descriptive proxies [D], not ground-truth latency, because no capture contains a truth-grade vertical-velocity signal. A commercial reference vario (baseline B03) was planned but remained unavailable [U].

3.4 Filters under test

Twenty-nine configurations in five families are compared against three baselines: B00 raw differencing; B01 least-squares slope over a 1 s window; B02 a literature-style project Kalman filter (q_h=0.05, q_v=0.5, auto-r). Family A: window smoothers (SMA, EMA, Holt, median, Hampel, robust EMA). Family B: frequency-domain low-pass (1-pole, 2nd-order Butterworth, 2nd-order Bessel, Savitzky-Golay). Family C: alpha-beta and alpha-beta-gamma trackers. Family D: Kalman variants (CV, CA, adaptive-R, innovation-gated). Family S: robust/gated estimators (S1 robust-adaptive, S1P persistence, S2 dual-timescale, S3 acceleration-aware, S4 fast/slow gated, S5 light-gate). All filters run from a single command-line entry point; every configuration and run is reproducible from the repository.

3.5 Tuning, validation, and metrics

Per-filter tuning was fit on training sessions only and reported against held-out validation sessions (whole-session separation). Static metrics: detrended altitude noise, drift range and Allan deviation, and Kalman-filter rest vertical-velocity RMS. Motion metrics: onset/release latency proxies, false-latch rate, and event-level audio behaviour (Section 4.4). Rate comparisons use physically equivalent parameterizations wherever possible (e.g. time-constant-matched EMA, bandwidth-matched low-pass).

4. Results

4.1 Static noise and drift: the draft effect

Table 2 condenses the cross-generation static characterization (results/21_static_compare.csv). Two findings stand out. First, the pressure ADC word dominates: per-sample pressure noise corresponds to 0.40–0.46 m detrended altitude, while the temperature ADC word contributes 0.2–0.5 Pa — an earlier hypothesis that temperature-ADC noise materially limits altitude was refuted [M]. Second, the quiet-house capture t008 drifts 1.25–1.38 m over 60 s, essentially matching the office statics t005/t006 (1.19–1.65 m), while the t007 office static — recorded beside an open-window draught — drifts 9.2–9.8 m, roughly 7x more. The Kalman rest vertical-velocity RMS follows the same split: 0.10–0.11 m/s on the quiet captures versus 0.71–0.80 m/s on the draughty one. The dominant limiter on practical vario noise is ambient pressure disturbance, not the sensor [M].

Table 2. *Static characterization across capture generations (per-sid ranges, both barometers; exploratory KF q=10 with per-capture r). Source: results/21_static_compare.csv [M].*

Capture	OSR/rate	r_est (m²)	Alt. noise (m)	Drift range (m)	ADEV 1s/10s (m)	KF rest v-RMS (m/s)
t005 office	4096 @ 45 Hz	0.0328–0.0367	0.43–0.44	1.34–1.65	0.39–0.40	0.120–0.131
t006 office	2048 @ 90 Hz	0.0269–0.0312	0.41–0.42	1.19–1.28	0.27–0.29	0.104–0.107
t007 office (draft)	2048 @ 90 Hz +IMU	0.0379–0.0463	0.44–0.46	9.21–9.76	0.41–0.51	0.713–0.796
t008 quiet house	2048 @ 90 Hz +IMU	0.0261–0.0326	0.40–0.42	1.25–1.38	0.27–0.30	0.105–0.112

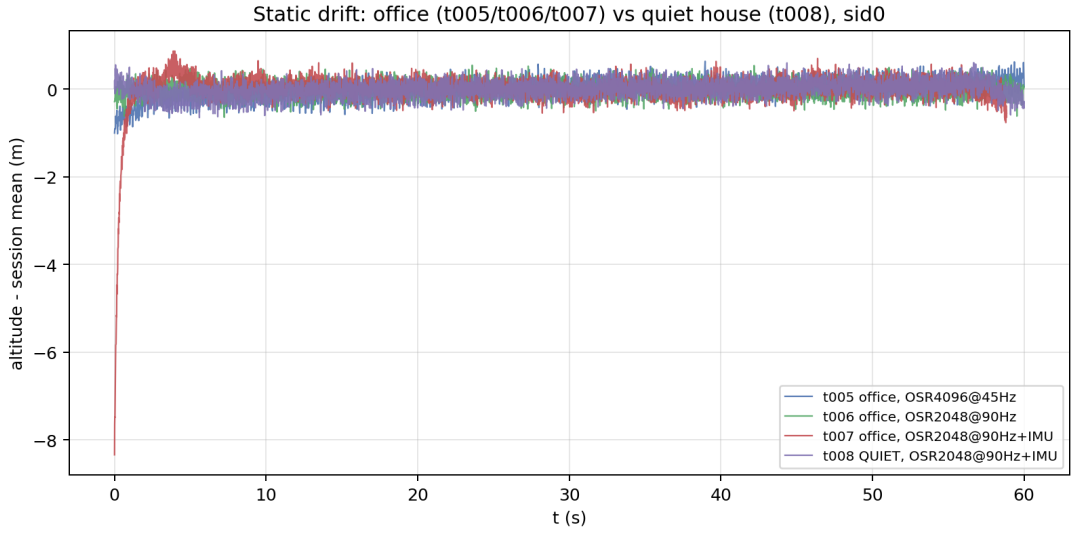


Figure 1. *Detrended altitude over the four 60 s statics. t007 (orange) shows the 7x larger draft-driven excursion; t005/t006/t008 overlay within ~0.4 m of each other.*

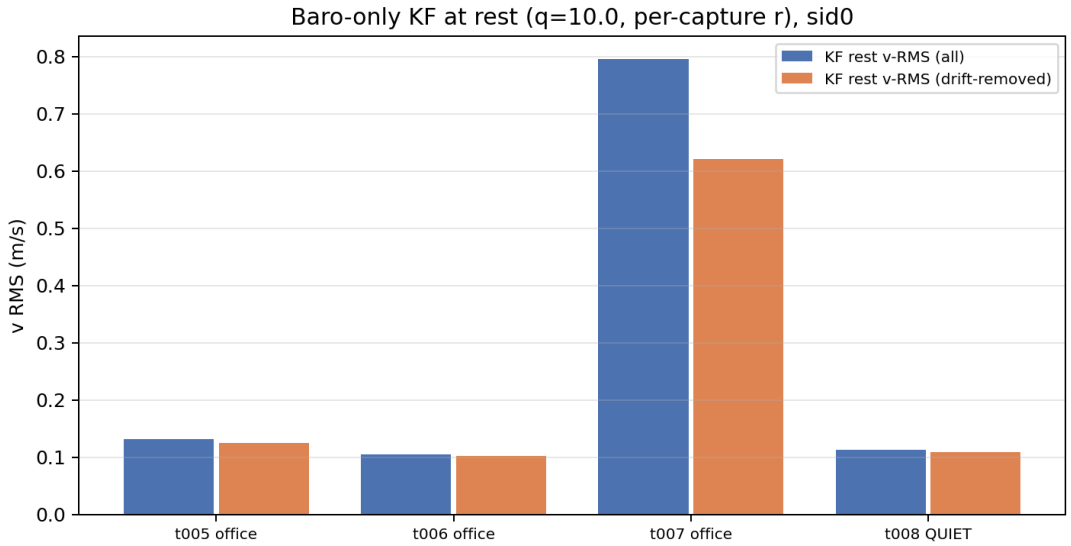


Figure 2. *Kalman vertical velocity at rest: quiet captures sit at the 0.10–0.11 m/s floor while the draughty t007 sits at 0.71–0.80 m/s — the environment, not the estimator, sets the floor.*

Allan deviation (Figure 3) is consistent with this reading: every capture shares a short-tau floor of 0.25–0.35 m, and only t007 rises with averaging time, the signature of a slowly varying disturbance (draft) rather than sensor white noise [M][D].

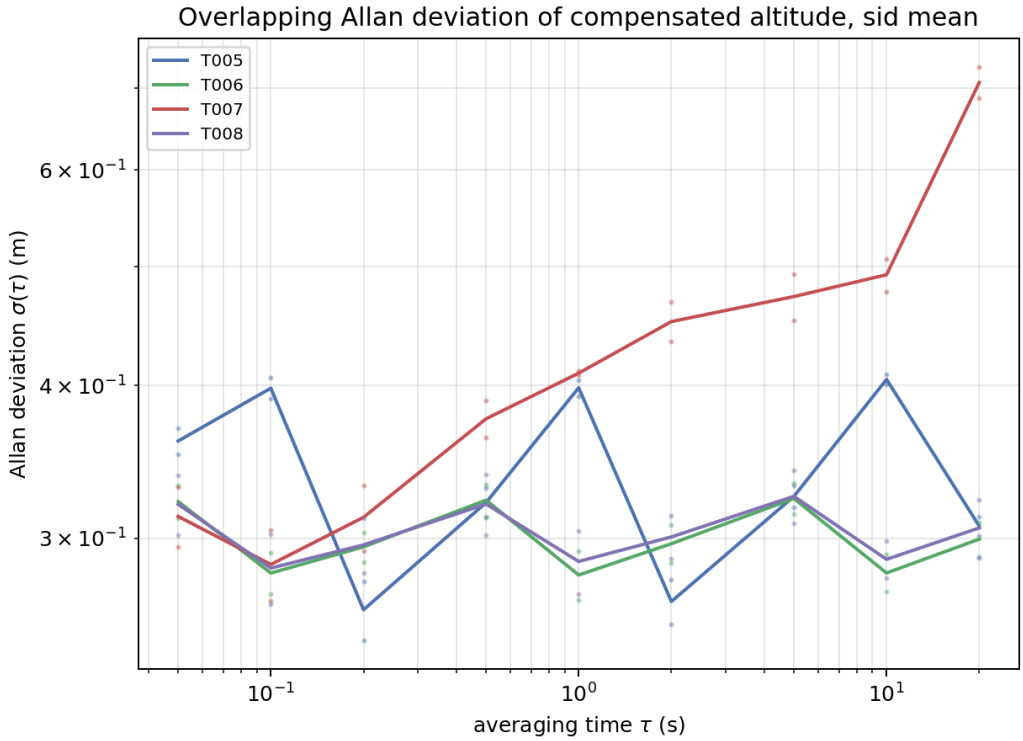


Figure 3. *Allan deviation of detrended altitude. Quiet statics are flat or falling with tau; the draughty t007 rises, indicating non-stationary environmental disturbance.*

4.2 IMU health at rest

The MPU6050 is genuine but consumer-grade: at rest it reports |a| = 10.055 m/s² (10.077 m/s² in t008) against 9.807 expected — a +2.5–2.8% clone-scale bias — while gyro offset norms (51.7–55.8 dps across sessions) are large but stable, and 1-sigma accel noise at rest is 0.013 m/s² [M]. Rest accel bias was estimated to 0.0014 m/s² residual norm by the fusion estimator under a rest gate. Madgwick gravity-direction error at rest is 0.011 rad, rising to 0.14 rad on the smooth elevator and 3.4–10.3 rad-equivalent residuals (as gravity-norm deviation) under tumble (Section 4.5).

4.3 Filter benchmark

Across both rates, the 1-pole and 2nd-order low-pass family is uniformly poor at rest (0.34–5.18 m/s RMS; the 2nd-order filters ring after transients), window smoothers trade noise for heavy latency, and the recursive estimators dominate. Table 3 gives the rate comparison on the static metric for representative filters (results/06_rate_comparison.csv); doubling rate with physically equivalent parameterization leaves the ordering unchanged and marginally improves the Kalman rest floor (0.101 to 0.087 m/s for the CV filter).

Table 3. *Static rest vertical-velocity RMS, representative filters (t001 statics; 06_rate_comparison.csv) [M].*

Filter	Family	Rest v-RMS 45 Hz (m/s)	Rest v-RMS 90 Hz (m/s)
B00 diff	baseline	9.02	21.5
B01 regression (1 s)	baseline	0.159	0.155
A01 SMA (1 s)	A	0.250	0.396
A02 EMA (tau=1 s)	A	0.190	0.175
B02 Butterworth-2	B	2.54	3.73
C01 alpha-beta	C	0.026	0.025
C03 CV tracker	C	0.047	0.056
D01 Kalman CV	D	0.101	0.087
D04 Kalman adapt-R	D	0.122	0.097

The retained Phase-4 candidates are summarized in Table 4 (results/12_estimator_summary.csv, 12_cpu.csv). Two defects shape the shortlist. First, the dual-timescale S2 froze permanently on its first sustained event: root cause [M] was the replace step writing the fast-track state from a prediction that itself consumed the fast-track state (self-reference), fixed by saturating the correction to a bounded deviation from prediction; S2 was disqualified. Second, S1 — the best rest performer — misses 18.9% of validation lifts under the common audio policy (Section 4.4). S5 matches S1's rest accuracy at 2.7x lower CPU; S4 is the fastest on release; C03 is the cheapest but false-latches and has a validation missed-lift rate of 2.7%.

Table 4. *Retained candidates: rest accuracy (training statics), derived onset/release latency proxies (validation), and measured CPU cost. Negative onset proxy for D01 is a proxy artifact (output leads the derived reference).*

Estimator	Rest RMS train (m/s)	Onset proxy val (s)	Release proxy val (s)	CPU (us/sample)
C03 CV tracker	0.048	1.29	2.18	7.1
D01 Kalman CV	0.096	-2.35	2.11	16.0
S1 Robust-adaptive	0.045	0.14	4.95	50.7

Estimator	Rest RMS train (m/s)	Onset proxy val (s)	Release proxy val (s)	CPU (us/sample)
S1P Persistence	0.045	0.14	1.90	50.4
S4 Fast/slow gated	0.085	0.04	1.43	19.0
S5 Light-gate	0.046	0.12	4.23	18.4

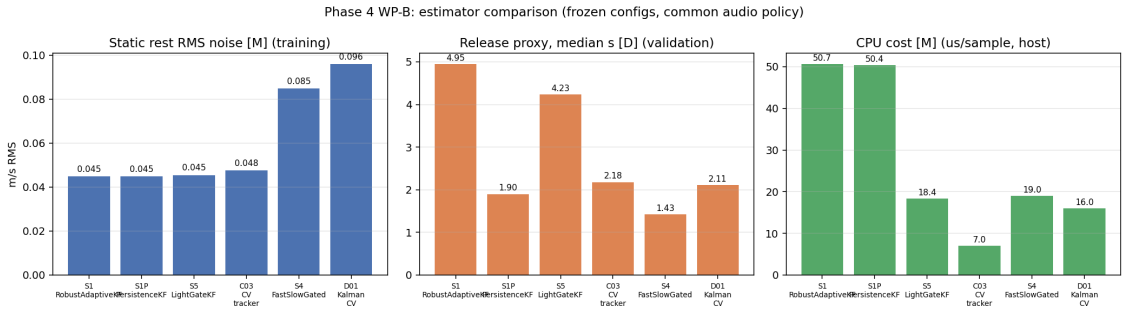


Figure 4. *Phase-4 candidates on rest-RMS versus release proxy: the accuracy/latency/CPU trade space. S2 excluded (freeze defect).*

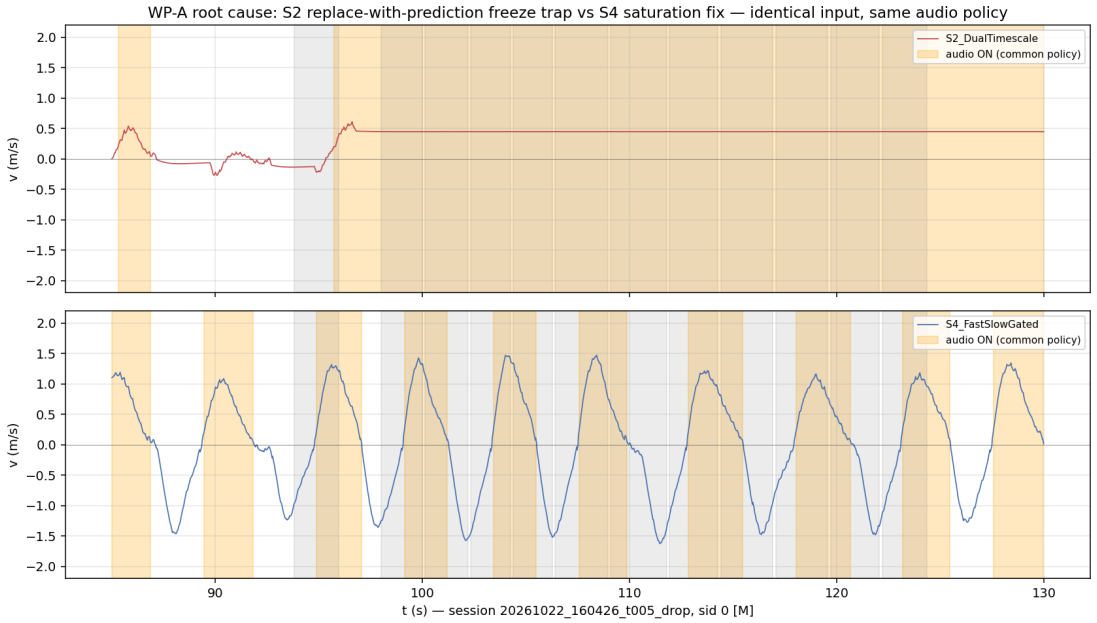


Figure 5. *The S2 freeze defect: fast-track state locked at its pre-event value while the slow track continued — traced to a self-referencing replace step.*

4.4 Audio behaviour: onset, exit-from-lift, and defects

Each estimator drives a common audio state machine with per-filter tuned thresholds (lift-on, lift-off, strong-lift), hysteresis and a persistence gate; the evaluation counts missed lifts, false-on time, onset delay and exit delay per event against derived references (results/13_policy_comparison.csv, validation split, tuned+safety variant, Table 5). The exit-from-lift requirement is explicitly measured: XCSoar-style conventions expect prompt silencing when lift ends [10], and continuous-beeping suppression is part of the policy score. S1/S1P pay for their rest accuracy with the worst lift detection (18.9–21.6% missed) — an unacceptable defect for a soaring instrument — while S4/S5/D01 detect all validation lifts. Transient stress tests (results/08_audio_failures.csv) further show estimators emitting extreme outputs after rejected transients (up to 16.2 m/s for S3 on the 90 Hz drop capture), which the audio persistence gate is designed to absorb.

Table 5. *Audio-policy evaluation, validation split. Missed-lift defect of S1/S1P is the key discriminator; false-on rates are from the training split (statics).*

Estimator	Missed-lift rate	Onset delay med (s)	Exit delay med (s)	False-on per hr (train)
C03 CV tracker	0.027	0.77	0.82	0
D01 Kalman CV	0.0	0.70	0.79	60
S1 Robust-adaptive	0.189	1.54	1.00	60
S1P Persistence	0.216	1.42	1.08	60
S4 Fast/slow gated	0.0	1.08	1.44	60
S5 Light-gate	0.0	0.90	0.98	60

4.5 Real-IMU fusion

Fusion was validated against the real t007/t008 captures (results/17_real_imu_fusion.csv, 22_t008_fusion.csv): Madgwick attitude at beta=0.3 from 200 Hz IMU samples, accel bias estimated only under a rest gate, Kalman core as in Section 2.2. At rest the fusion output is numerically identical to baro-only (t008: 0.1085 vs 0.1086 m/s; zero reinitializations) [M] — the IMU adds nothing when the rig is still, as it should. On the elevator the two are within 4% (1.39 vs 1.33 m/s session RMS). Under tumble, fusion is 1.4–2.2x worse than baro-only (Table 6): the attitude error, visible as gravity-residual norm 6.5–10.3 m/s² during drop/toss, injects false vertical acceleration. The bottleneck is therefore the 6-DOF attitude estimator without magnetometer — not the Kalman core [M][D].

Table 6. *Real-IMU fusion vs baro-only (sid 0; sid 1 within 5%). Fusion is neutral at rest, equal on smooth motion, and harmful under tumble — the attitude-limited regime.*

Session	Rest/behaviour	v-RMS baro-only (m/s)	v-RMS fusion (m/s)	Gravity residual (m/s²)
t008 quiet static	rest	0.1085	0.1086	0.011 rad
t007 bench static	rest (drafty)	0.756	0.756	0.011 rad
t007 elevator 2x	smooth motion	1.334	1.393	0.137 rad
t007 drop	tumble	1.245	2.077	6.53
t007 drop2	tumble	1.511	2.678	8.21
t007 toss	tumble	1.361	2.972	10.29
t007 ControledMove2	jerk + tumble	1.685	2.205	3.41

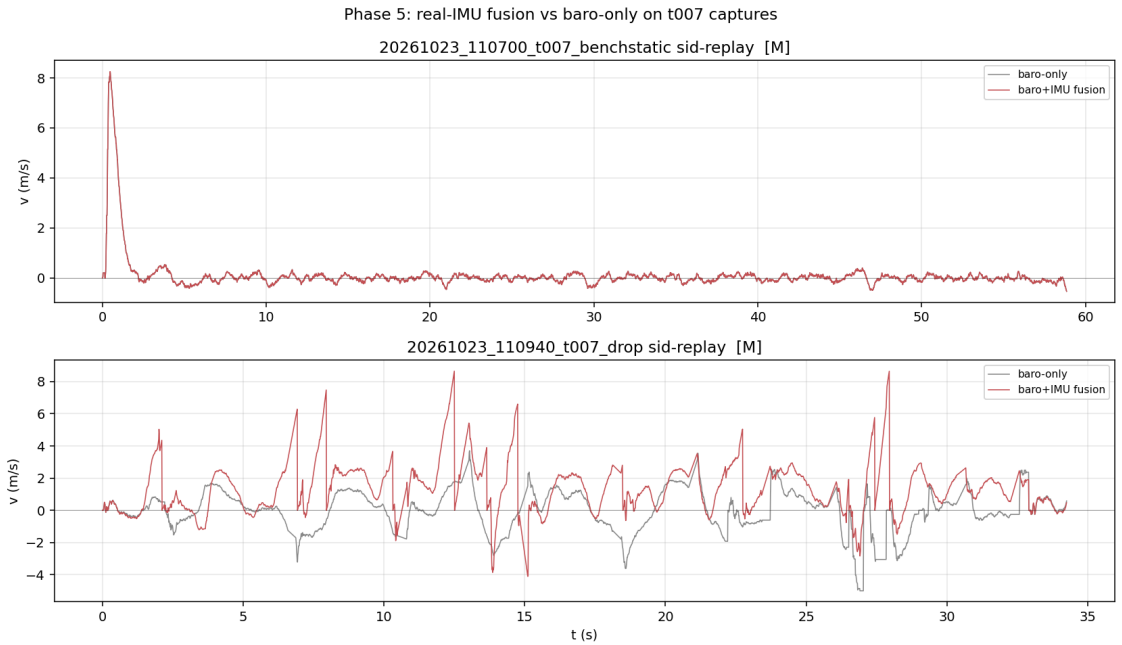


Figure 6. *Fusion versus baro-only on the t007 drop and elevator: the fused track diverges exactly where attitude error grows.*

4.6 Portability and fault handling

The Kalman core is portable: a C99 implementation matches the Python reference to 2.0e-08 m/s worst-case across 12 irregular-sampling, spike, gap and reversal vectors in double precision, and passes looser float tolerances; it compiles for the Xtensa ESP32-S3 target (results/15_portable_equivalence.csv) [M]. Fault injection (results/09_fault_recovery.csv) validates innovation gating at 3.5 sigma plus a loss-of-track reinitializer (8 consecutive >5 sigma innovations trigger reacquisition): after a sustained sensor fault the estimator recovers within one event window without latching. The shortest-arc attitude warm start (quaternion from accelerometer tilt) shipped after closed-loop replays showed cold-start attitude transients contaminating the first seconds of fusion; trigonometric forms were abandoned as convention-treacherous on the X-up rig in favour of a vector form.

4.7 Simulated fusion ceiling

To separate architecture from hardware, the identical fusion pipeline was run on simulated streams with known attitude (results/14_fusion_simulated.csv) [S]. With attitude given, fusion cuts climb-tracking RMS error from 0.499 to 0.0095 m/s (50x), coasts through a 1 s barometer blackout at 0.034 m/s RMS versus 1.70 m/s baro-only, and bounds a single-sample 8 m pressure spike to 1.50 m/s peak velocity error. The architecture is sound; the hardware attitude estimate is what stands between the bench rig and these numbers [S][D].

5. Discussion

Three measured results reorganize the design space. First, the quiet-static floor of 0.10–0.11 m/s rest vertical-velocity RMS — reached by a plain CV Kalman with per-capture r — is already at the level pilots describe as 'quiet', and it was reached without any fusion. The 7x degradation caused by a room draft shows the deployment environment dominates; a filter tuned on a draughty bench would be over-smoothed for the air. Second, IMU fusion as implemented is an attitude problem in disguise: with a 6-DOF IMU and no magnetometer, gravity residuals during tumble exceed the signal by an order of magnitude, so fusion helps exactly where attitude is good (rest, smooth motion) and hurts where attitude fails. Confidence-gated fusion — applying IMU propagation only when attitude error is bounded — is the natural next step, and the simulation ceiling quantifies the payoff (Section 4.7). Third, accuracy and audio behaviour diverge: the estimator with the best rest RMS misses nearly one lift in five, so filter selection must be driven by the policy metrics of Table 5, not by RMS alone.

The XCTracer-style architecture (fast IMU-predict/baro-correct Kalman with a slow baro-only confidence channel) remains a candidate, supported by the fusion results here, but its implementation details are not assumed without evidence and no estimator is frozen: further hardware testing and comparative measurements are planned before selection, per the programme constraint.

6. Limitations

• All latency figures are derived proxies against derived event references [D]; no capture contains ground-truth vertical velocity, and the planned commercial reference vario (B03) was unavailable [U].

• The IMU is a genuine but low-grade MPU6050 with +2.5–2.8% accel scale bias and 52–56 dps gyro offsets; results characterize this unit, and a different IMU would shift the fusion conclusions.

• Both barometers are clones from one supplier batch; clone-to-clone variance beyond this pair is unmeasured.

• Bench motion (drops, tosses, elevator) is a proxy for thermal-lift flying; no real in-flight captures exist yet, and drop/toss dynamics exceed any soaring regime, which is why fusion looks worse on them than it likely would be in air [H].

• The KF q value used in the static characterization (q=10) is exploratory; q discrimination requires motion-session tuning and held-out validation, not statics.

• The USB capture path itself was implicated in a host-side freeze during long sessions; the MCU-side root cause is unverified [U].

• Audio-policy thresholds were tuned on bench events; pilot-in-the-loop validation is outstanding [U].

7. Conclusions

On committed raw captures and a causal replay engine, this bench programme measured, rather than assumed, the noise floor, the filter trade space, the audio behaviour, and the real-IMU fusion value of a dual-barometer ESP32-S3 variometer. A plain CV Kalman filter reaches a 0.10–0.11 m/s rest floor in a quiet environment; ambient drafts, not the MS5611 clones, set the practical limit. IMU fusion is neutral at rest and on smooth motion but attitude-limited under tumble, with a quantified 50x ceiling if attitude is known. Three stacked-filter candidates (S5 light-gate, S4 fast/slow gated, D01 CV Kalman as the simple reference) are retained with measured trade-offs; S1/S1P are retained only if their missed-lift defect is resolved. All data, scripts, and result tables are committed and reproducible from the repository. The next engineering step is adaptive attitude-trust gating of the IMU fusion path, validated on controlled-motion captures, followed by real thermal-lift flight captures for final estimator selection.

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Reproducibility: all captures in analysis/data/, all metrics in filter_benchmark/results/.csv, all figures in filter_benchmark/results/*.png, and the generators of both this paper and the companion engineering report in docs/pdf/. Build with python docs/pdf/build_technical_paper.py.*